

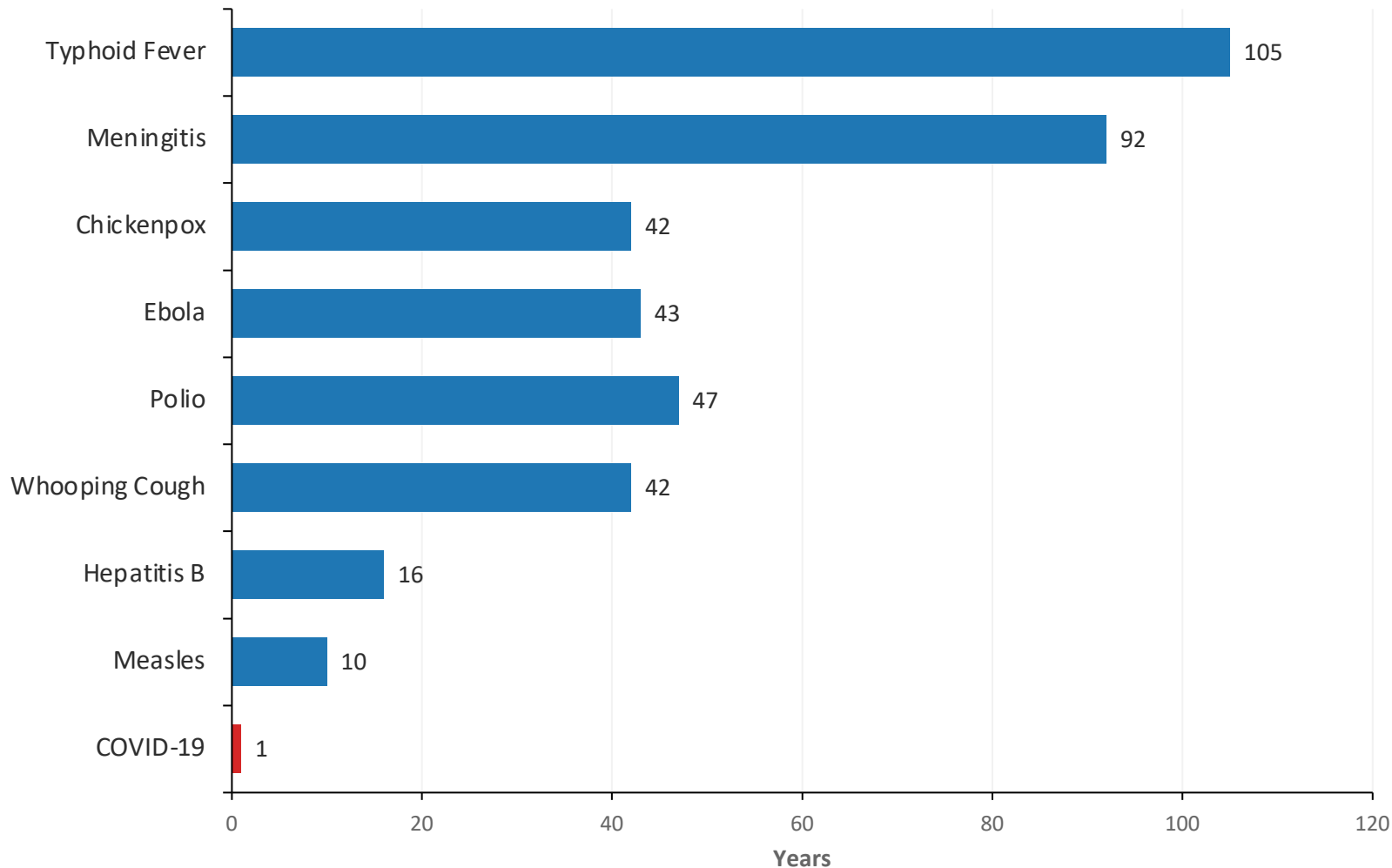
The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines have transformed public health
across two centuries (1796 – 2020)

Source: Our World in Data | For public health professionals & policy analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



COVID-19: < 1 Year

mRNA technology & genomic sequencing enabled the fastest vaccine in history

Typhoid: 105 Years

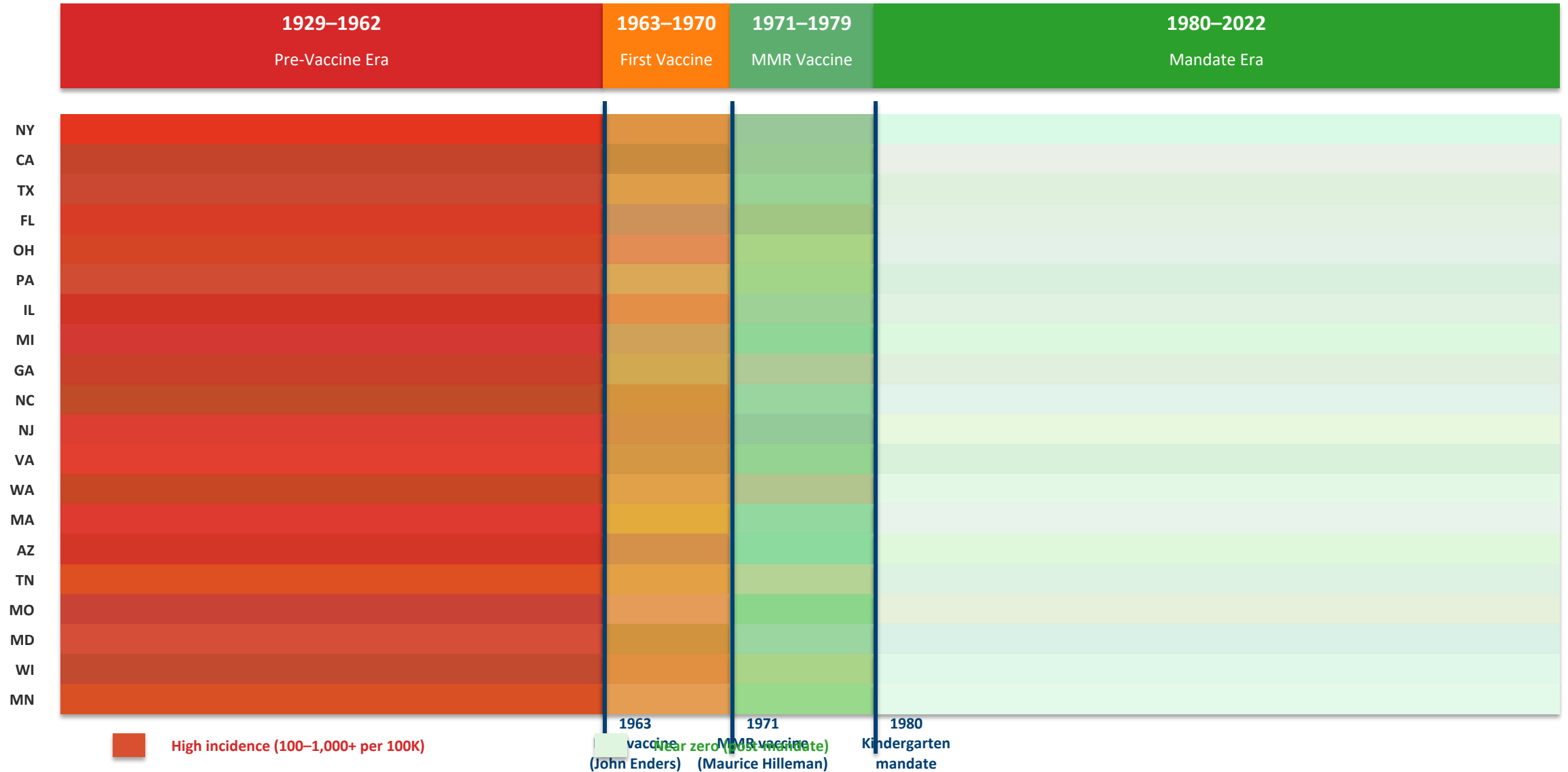
Early vaccines took decades — scientists first needed to establish germ theory

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

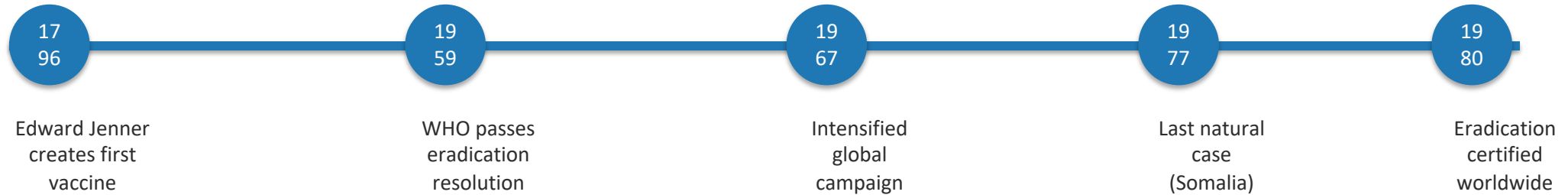
Disease	Pre-Vaccine Cases/M/yr	Post-Vaccine Cases/M/yr	Case Reduction	Pre-Vaccine Deaths/M/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute Polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Green rows = 100% elimination of cases and deaths | Source: Roush & Murphy (2007)

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Smallpox: The Only Human Disease Eradicated Through Vaccination



Why Was Smallpox Uniquely Suited for Eradication?

- ✓ The virus infected only humans — no animal reservoir
- ✓ Symptoms were clearly distinguishable
- ✓ The vaccine was cheap and highly effective
- ✓ Global coordination through the WHO campaign

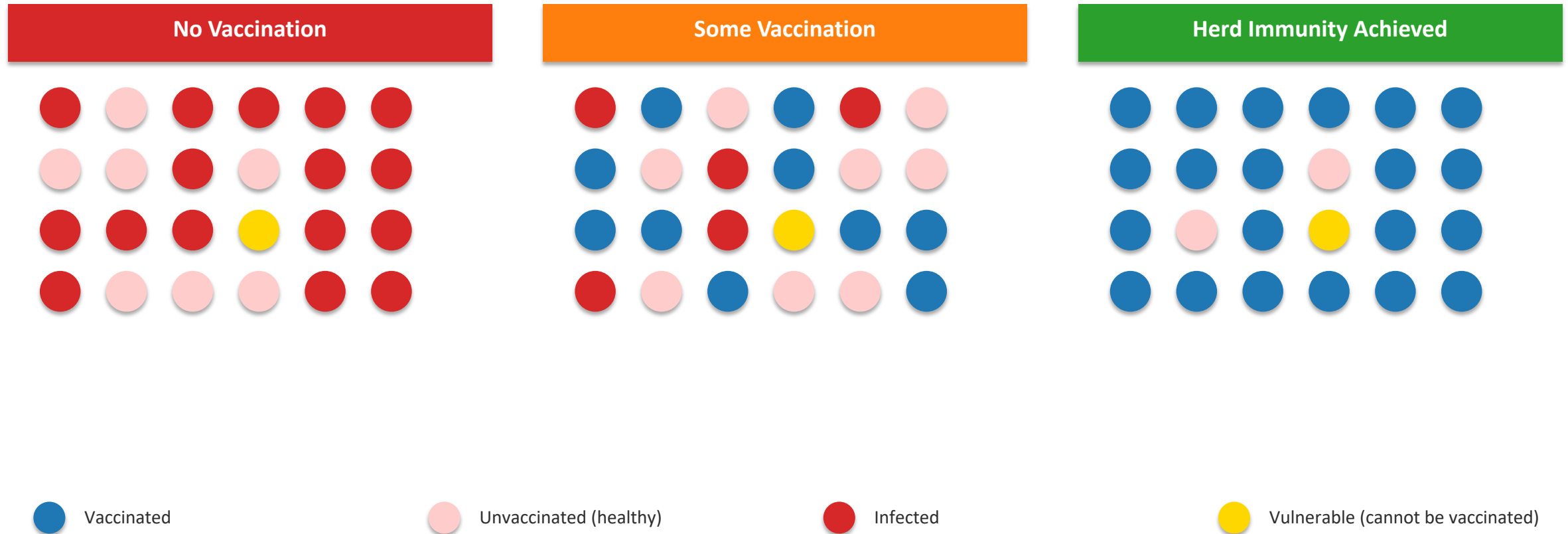
The Greatest Public Health Achievement

Declared eradicated worldwide

1980

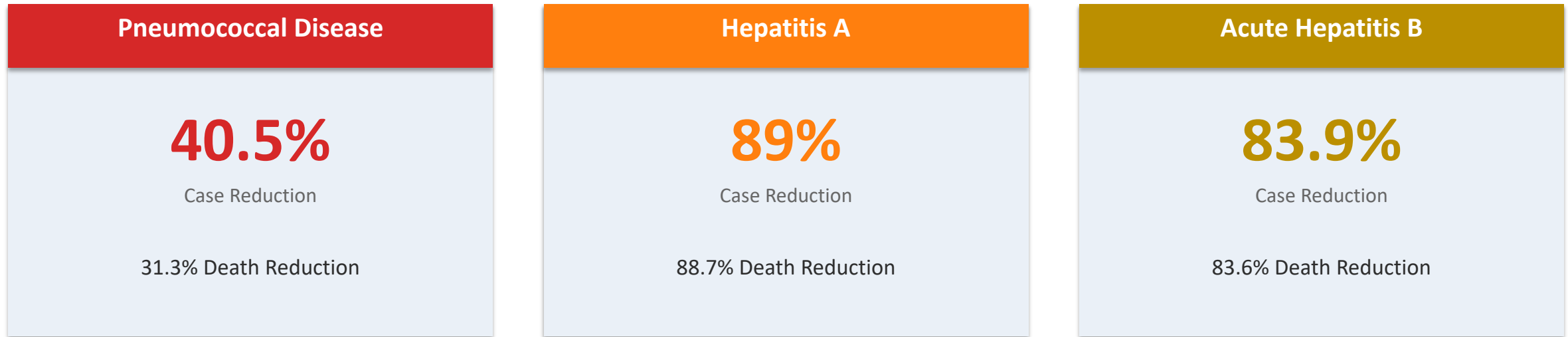
100% case & death reduction in the US

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



When enough people are vaccinated, pathogens cannot spread effectively — shielding newborns, immunocompromised individuals, and cancer patients on chemotherapy who cannot be vaccinated.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Key Barriers to Closing the Gap

Access Gaps: Not everyone has access to vaccines globally — supply shortages put children at risk

Trust Challenges: Poor scientific understanding can reduce public confidence and uptake

Vaccine Limitations: Some vaccines provide incomplete protection, requiring improved formulations

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

100% Elimination

Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US.

Timeline Collapsed

Development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology.

Smallpox Eradicated

Smallpox remains the only human disease eradicated through vaccination, certified in 1980.

Herd Immunity

Herd immunity protects entire communities — shielding newborns, immunocompromised individuals, and cancer patients.

Gaps Persist

Coverage gaps persist for pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%).

Call to Action: Strengthen global health systems — improve vaccine access, expand coverage, invest in R&D, and rebuild public trust.